Kaylee Pascarella

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Education

Baylor University — *Waco, TX*
Ph.D. in Statistics — *August 2026*

Dissertation: *Bayesian Frameworks for Composite Strategy Estimands and Pharmacokinetic AUC Estimation*
Advisor: John W. Seaman Jr.

Baylor University — *Waco, TX*
M.S. in Statistics, GPA 3.87 — *December 2023*

Baylor University — *Waco, TX*
B.S. in Statistics, GPA 3.92, Magna Cum Laude — *May 2022*

Research Experience

Graduate Researcher, Department of Statistical Science — August 2023 – August 2026
Baylor University — *Waco, TX*

- Developed two Bayesian estimators for composite strategy estimands with distributional failure value definitions under the ICH E9(R1) framework: a two-stage estimator propagating failure value uncertainty through Bayesian multiple imputation, and a parametric joint model estimated by Bayesian G-computation
- Designed and executed a simulation study spanning more than 120 scenarios at 5,000 replications each, benchmarking both estimators against four frequentist alternatives and producing method selection and sample size guidance for practitioners
- Constructed Bayesian models for area under the concentration curve (AUC) estimation in preclinical serial sacrifice pharmacokinetic studies
- Implemented all models in Stan and JAGS with reproducible, version-controlled R workflows, including prior predictive checks and prior sensitivity analyses

Data Science Fellow, MoWaTER Program — May – July 2022
Baylor University and Colorado School of Mines (NSF-supported) — *Waco, TX*

- Analyzed sensor data from a mobile direct potable reuse water treatment system spanning 71 operating sessions, characterizing normal operating conditions across processes
- Applied functional data analysis, registering operating cycles via cubic spline smoothing and using magnitude and variance outlyingness metrics with bagplots to isolate candidate equipment faults
- Evaluated clustering approaches for automated fault detection and co-authored the program technical brief

Professional Experience

Biometrics Intern, Late Respiratory and Immunology Statistics — June – August 2025
AstraZeneca — *Gaithersburg, MD*

- Investigated failure value selection strategies for the composite strategy estimand applied to continuous respiratory endpoints, characterizing the sensitivity of treatment effect estimates to clinically motivated threshold choices
- Built an interactive R Shiny application enabling real-time comparison of treatment effect estimates across competing failure value strategies
- Performed quality control checks of clinical tables and reports and produced data visualizations supporting team analytical deliverables
- Delivered an end-of-internship presentation to Late R&I statisticians summarizing methodological findings and practical recommendations

Biostatistics Intern — August 2024 – Present
Baylor Scott & White Health

- Contributed to statistical analysis plans and performed analyses for clinical research studies, including survival analysis, retrospective cohort and longitudinal EHR analyses, rank-based and categorical methods, hypothesis testing, and sample size determination (R, SAS)
- Produced statistical tables and figures and authored statistical methods and results sections, coordinating with study teams and clinical investigators through review cycles to finalize deliverables
- Communicated results to non-technical clinical audiences through written reports and ongoing consultation with medical residents and faculty

Statistical Consultant May – August 2024
Baylor University Statistical Consulting Center *Waco, TX*

- Applied Firth penalized logistic regression to predict historical lead presence across 14,000+ municipal water service lines, addressing complete separation that rendered conventional logistic regression estimates unstable
- Identified approximately 95% of unknown-material lines as not requiring replacement, narrowing the replacement candidate set for the client utility
- Scoped open-ended problems across clinical, public health, and environmental domains, translating research questions into statistical approaches and written deliverables for non-specialist audiences

Publications

Pascarella, K., & Seaman, J. W. "A Bayesian Approach to the Composite Strategy for Handling Intercurrent Events for Continuous Outcomes." Manuscript in preparation for the *Journal of Biopharmaceutical Statistics*.

Alnazeer, M., Fan, J., Giang, T., Euell, C., Lee, J., Izekor, B., Perez, C., Zamin, S., **Pascarella, K.**, Banchs, J., & Olsovsky, G. "Robotic Magnetic Navigation Versus Manual Catheter Ablation for Premature Ventricular Contractions: A Single-Center Retrospective Cohort Study." *Journal of Innovations in Cardiac Rhythm Management*, 2026.

Presentations

Pascarella, K. "Bayesian Approach to Composite Strategy for Handling Intercurrent Events for Continuous Outcomes." *Joint Statistical Meetings*, Boston, MA, August 2026. (Poster)

Pascarella, K. "Bayesian Approach to Composite Strategy for Handling Intercurrent Events for Continuous Outcomes." *Midwest Biopharmaceutical Statistics Workshop*, Carmel, IN, May 2026. (Poster)

Teaching

Supplemental Instructor August 2022 – May 2024
Baylor University *Waco, TX*

- Led weekly statistics review sessions for 30+ undergraduate students, communicating fundamental concepts through worked examples, guided practice, and one-on-one instruction

Service and Leadership

Paper Club Organizer, Department of Statistical Science August 2025 – August 2026
Baylor University *Waco, TX*

- Organized a monthly paper club for statistics Ph.D. students, handling logistics and leading discussions on methodology papers and professional development topics

Technical Skills

Statistical Methods: Bayesian inference and prior specification; ICH E9(R1) estimands and intercurrent event strategies; clinical trial and experimental design; sample size determination and power analysis; clinical trial simulation and operating characteristic evaluation; causal inference and G-computation; survival, longitudinal, and categorical data analysis; missing data under MAR and MNAR; multiple imputation; pharmacokinetic AUC estimation; functional data analysis

Software: R (Stan/rstan, JAGS, tidyverse, ggplot2, Shiny, RMarkdown/Quarto), SAS, Python, SQL, Git/GitHub, LaTeX

Deliverables: Statistical analysis plan contributions, statistical methods and results write-ups, tables and figures, reproducible reporting, investigator and client consultation

Selected Graduate Coursework

- Design of Experiments and Clinical Trials
- Bayesian Methods and Data Analysis
- Bayesian Theory
- Causal Inference
- Survival and Reliability Analysis
- Computational Statistics I and II
- Theory of Statistics I and II
- Large Sample Theory
- Analysis of Categorical Responses
- Multivariate and High-Dimensional Analysis
- Time Series Analysis
- Spatial Statistics